



FAPS

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Insulation systems and processes

Vorlesung Produktion elektrischer Motoren und Maschinen
Lecture Production of electric motors and machines

The content of the lecture unit is intended to provide a basic overview of the relevant insulation steps involved in the manufacture of an electric motor.

Insulating materials are used to provide potential separation within the coils and contribute significantly to the operation of the electrical machine through their use.

After attending the lecture unit "Insulation systems and processes" you should:

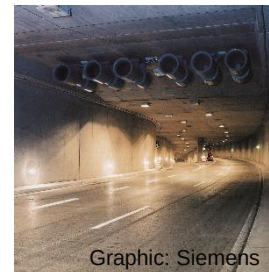
- know the essential properties of insulation materials and their processing,
- know application methods and the classification of primary insulation,
- understand the most important processing methods for applying secondary insulation and
- know the most important failure modes and relevant tests.



Graphic: Omron



Graphic: ZF Friedrichshafen



Graphic: Siemens



Graphic: Kuka

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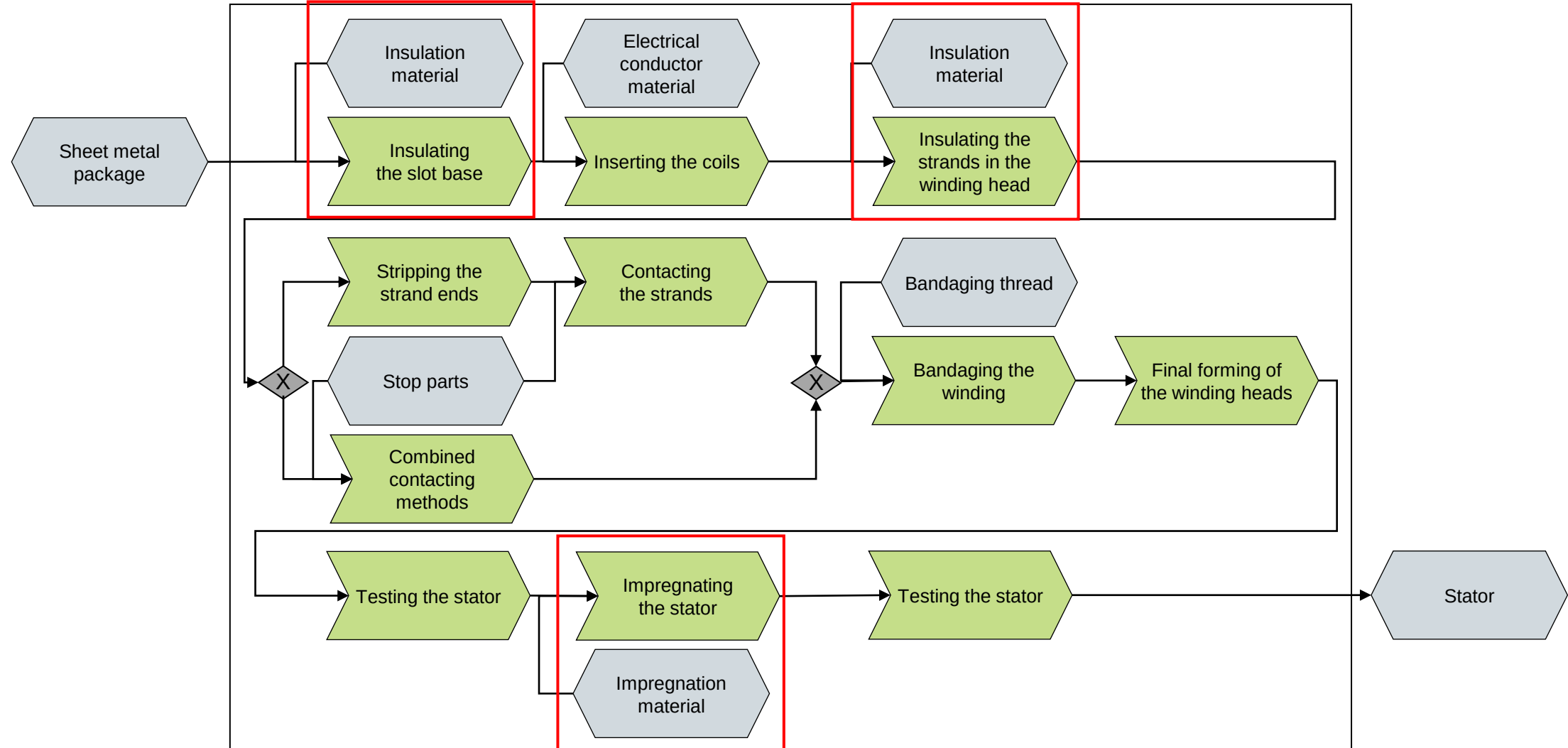
- **Basics of electrical insulation systems**
- Materials and types of insulation systems
- Process for applying the primary insulation
- Method for applying the secondary insulation
- Resin-based impregnation and potting processes
- Failure modes and testing technology



Graphic: azom.com

The production chain for the manufacture of a stator with a distributed winding of round wire essentially consists of nine steps.

Slide from Lec. 6



Insulation materials can be obtained both from natural as well as synthetic materials.

Electrical insulation

Definition: An insulator causes an almost complete potential separation of a conductor from its surroundings. Insulating materials are defined by their high specific resistance ($>10^7 \Omega$).

Materials

Ceramics

- Aluminum oxide
- Zirconium oxide



Glass

- Lead glass
- E-glass



Natural substances

- Rubber
- Mica

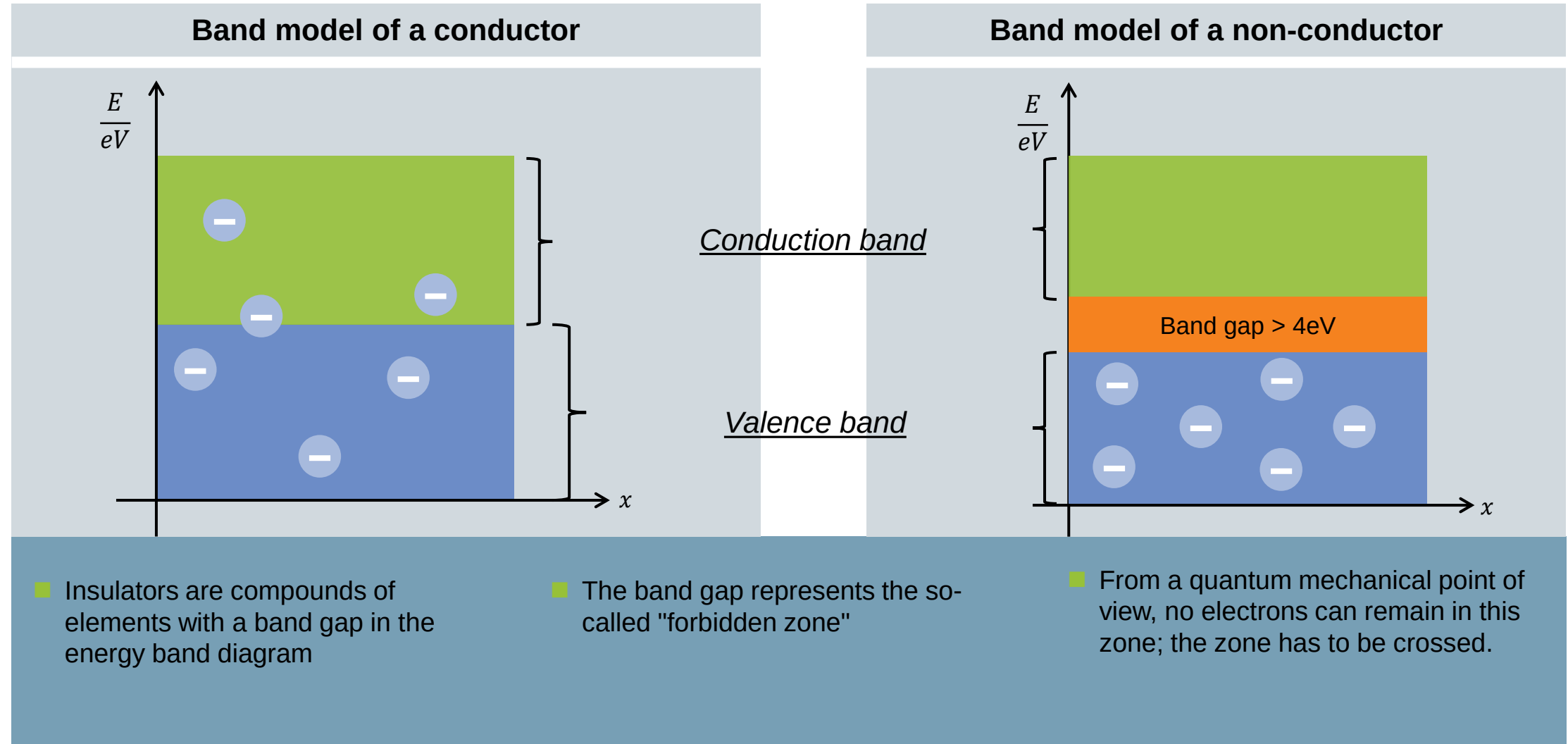


Polymers

- Thermosets
- Elastomers



Due to their special properties, insulators are the ideal materials for the electrical isolation of conductors from each other.

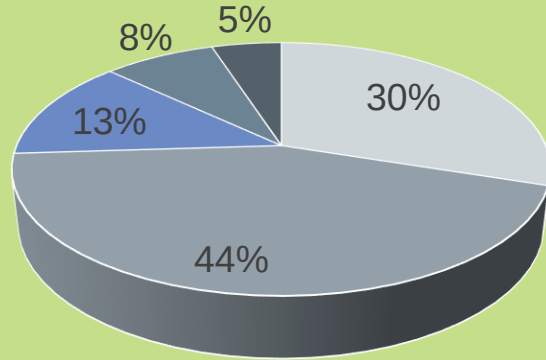


Insulation layers are an important lever in the further development of electrical machines in terms of output power and durability.

Role of insulation components in the further development of electrical machines

Relevance of insulation

Nearly 50% of motor failures are due to insulation failure

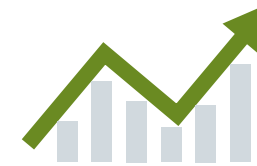


■ Overload ■ Insulation failure
 ■ Bearing damage ■ Rotor damage
 ■ Other

Source: Rockwell Automation



Power density



Production volume



Durability

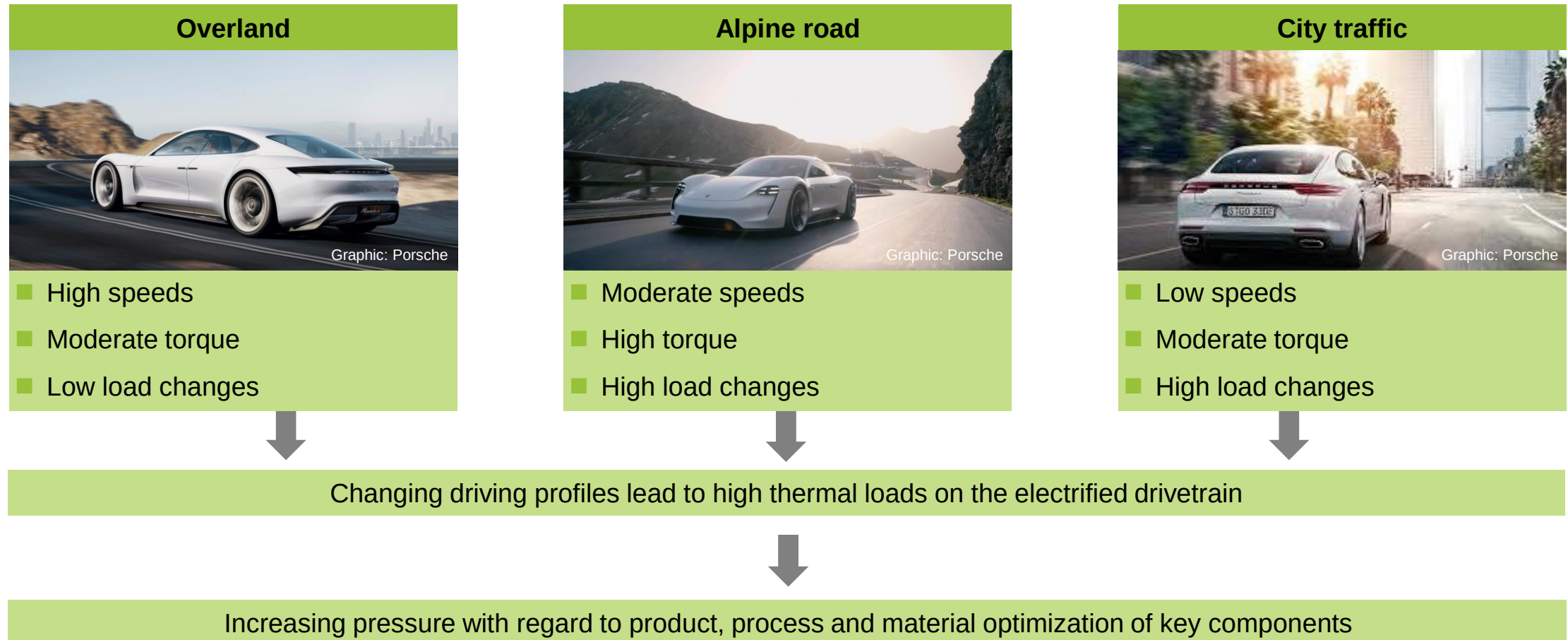
- Insulation layers form thermal bridge from wire to laminated core
- Thickness and material properties have a direct influence on power density and costs
- Production processes with increased output are necessary



Are existing systems up to the current challenges?

The requirements placed on electric traction drives in operation differ significantly from stationary industrial drives.

- Broad requirements profile for electric traction drives
- No acceptance of performance losses by the customer



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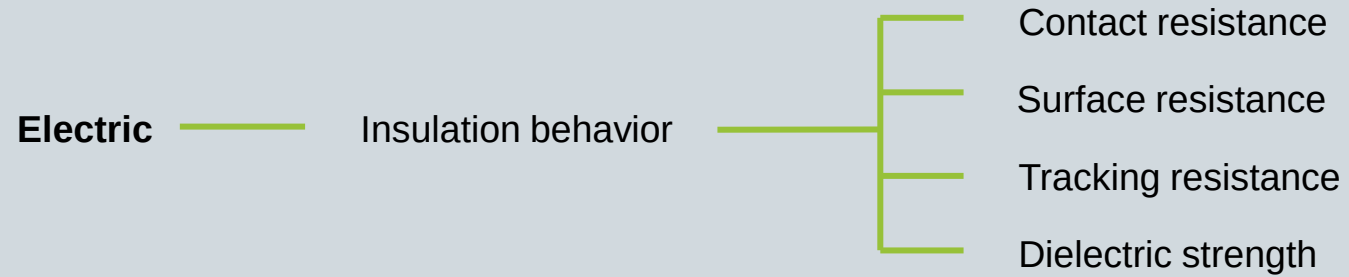
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Graphic: azom.com

Insulation materials can be classified according to various aspects.

Classification of electrical insulation



Thermal expansion
Thermal conductivity
Heat resistance
Heat resistance

Thermal



Insulators are primarily classified on the basis of their electrical properties.

Contact resistance	Surface resistance	Tracking resistance	Dielectric strength
■ The resistance to the flow of current through a material ■ often also simply "(electrical) resistance" ■ has the unit Ohm $1 \Omega = 1 \text{ V/A}$ ■ "Durchgangswiderstand"	■ The resistance to the flow of current between two electrodes placed on the surface of an insulator ■ has the unit Ohm $1 \Omega = 1 \text{ V/A}$ ■ "Oberflächenwiderstand"	■ Resistance of an insulator to currents that form on the surface ■ is determined with CTI value (voltage at which a leakage current occurs) ■ "Kriechstromfestigkeit"	■ Maximum electric field strength that may prevail in the material without a voltage breakdown (electric arc) occurring ■ Specification in kV/mm ■ "Durchschlagsfestigkeit"



Graphic: rm-prüftechnik

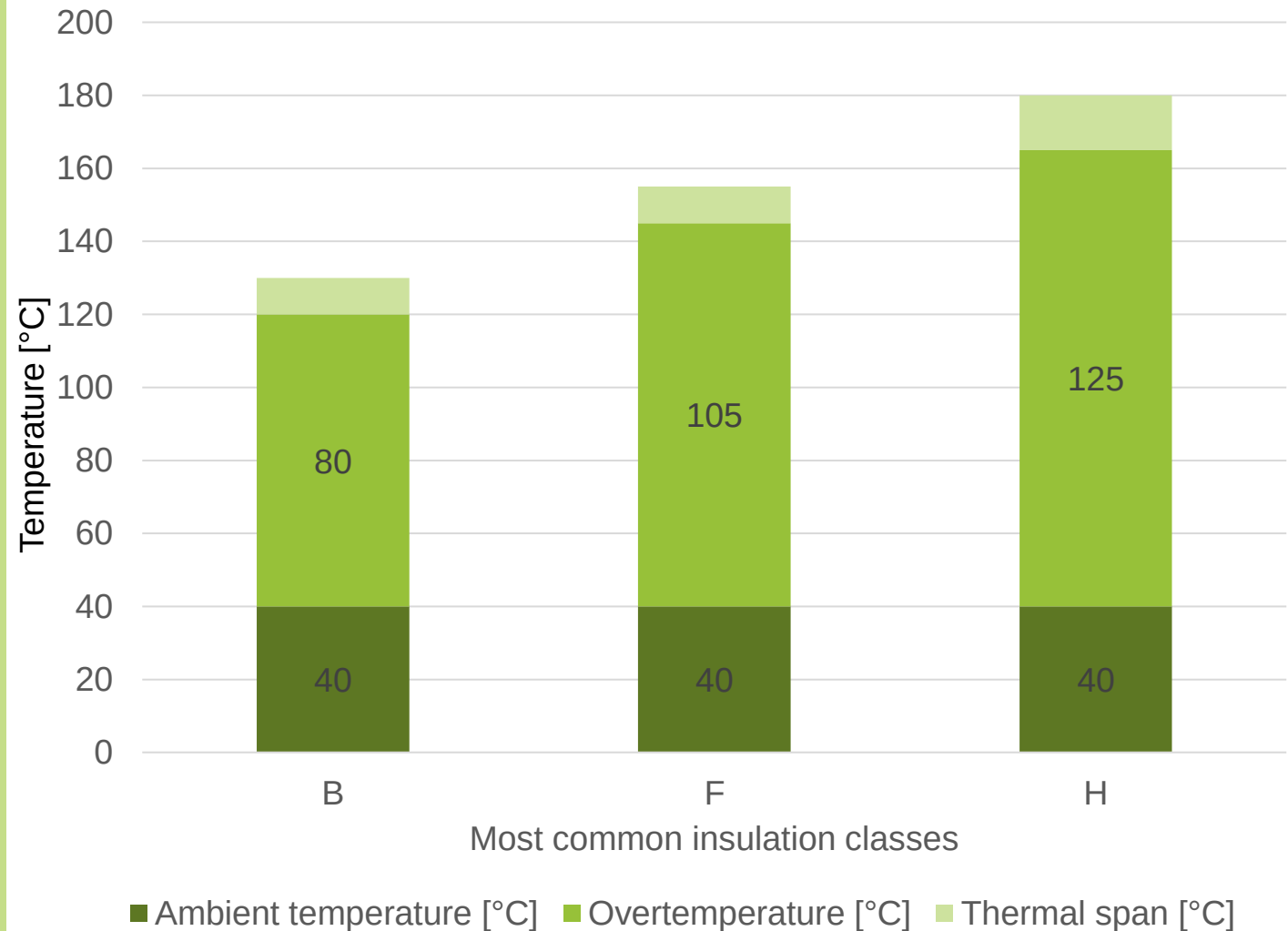
- Testing of substances and insulated stators is carried out using dedicated measuring methods and devices
- Test methods are often destructive and damage the insulation

The thermal classification of insulation materials is regulated by DIN EN 60034-1.

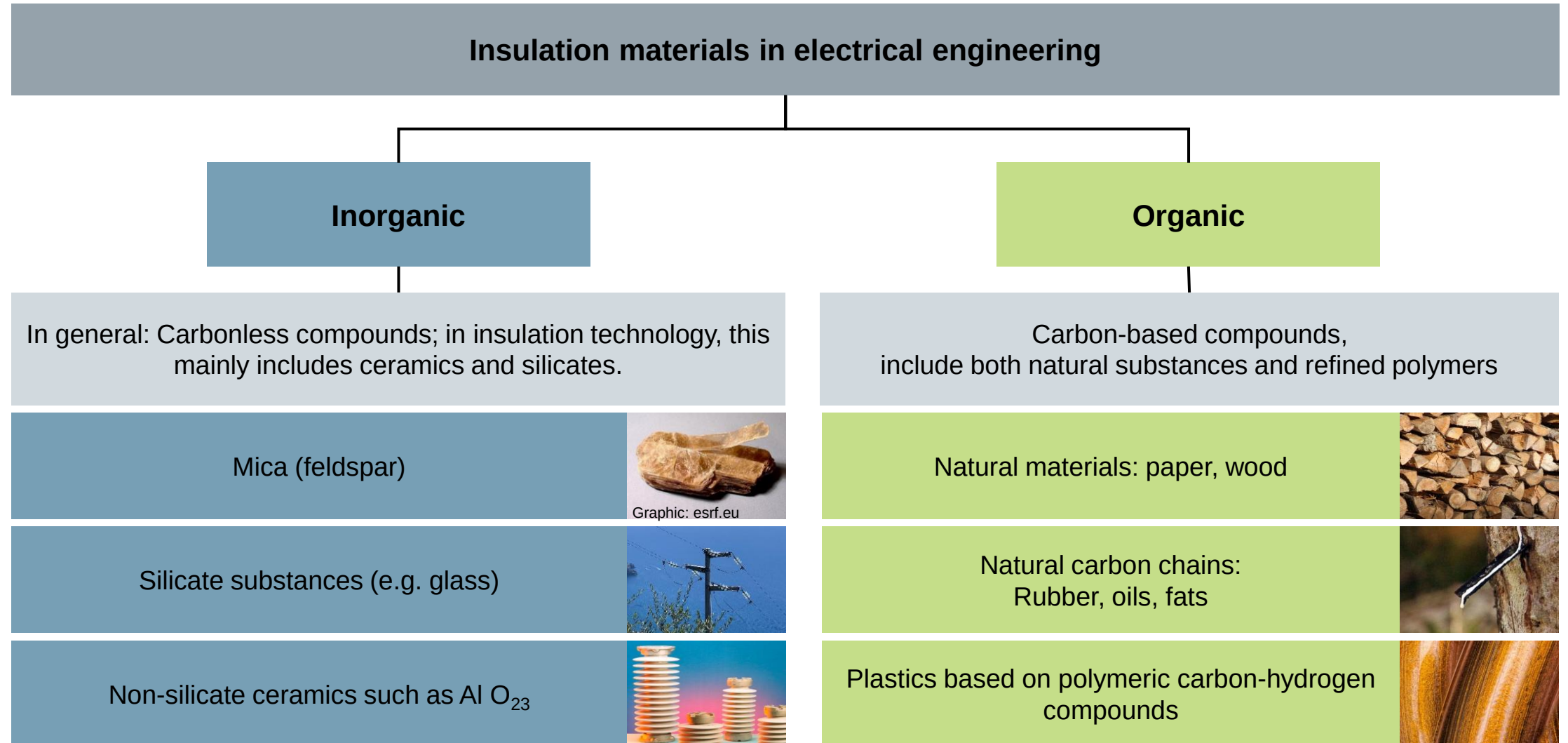
IEC (DIN EN) 60034-1:

- Classification of all insulation materials by temperature class, regardless of physical state and material type
- Within the limit temperature, insulating materials may lose a maximum of 50% of their physical properties after 20,000 operating hours
- Classification into ambient temperature, overtemperature and thermal margin (consideration of hotspots within system)

Insulation class	Maximum permissible temperature [°C]
Y	90
A	105
E	120
B	130
F	155
H	180
C	> 180



The most important insulators come from both inorganic and organic substances.

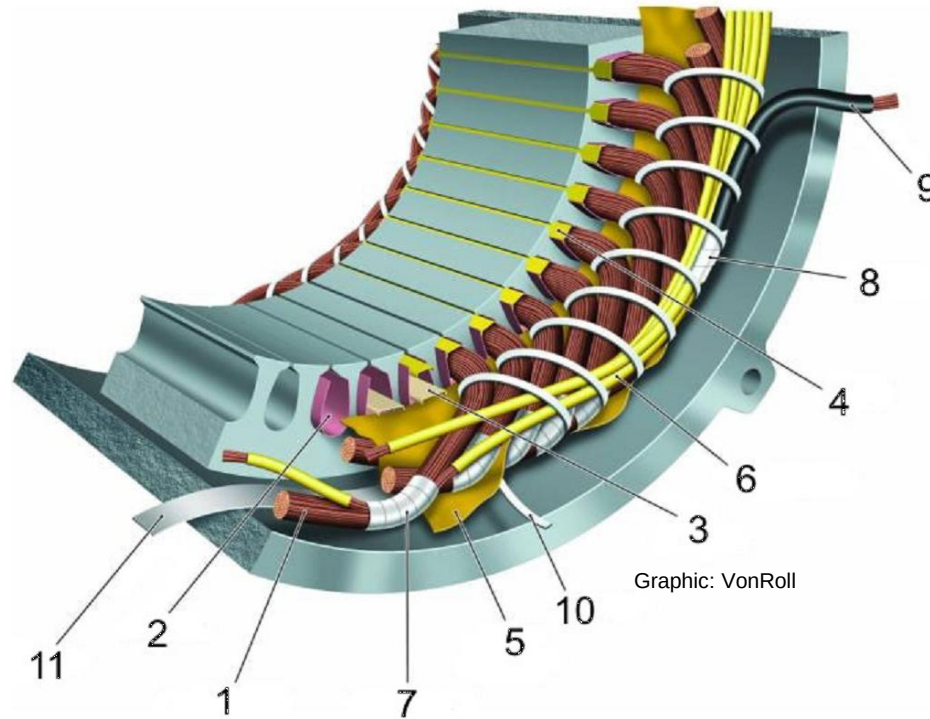


The most important insulators come from both inorganic and organic substances.

Property	Inorganic Insulators	Organic Insulators
Examples	Glass, ceramics, mica, porcelain	PVC, polyethylene, Teflon, paper, rubber
Specific resistance	10^{14} – $10^{23} \Omega \cdot \text{m}$	10^{11} – $10^{17} \Omega \cdot \text{m}$
Thermal resistance	500–1800 °C (depending on material)	80–300 °C (usually <200 °C)
Dielectric strength	20–60 kV/mm (typical)	15–105 kV/mm (typical)
Chemical resistance	Very high	Variable, often sensitive
Mechanical strength	High (often brittle)	Variable, often elastic
Processing/formability	More difficult, brittle	Very good, flexible, inexpensive
Typical applications	High-voltage insulators, furnace construction, capacitors, electron tubes	Cable insulation, circuit boards, connectors, electronic housings
Remarks	Extremely high insulation, heat and chemical resistance	Lightweight, flexible, versatile, but less temperature and aging resistant

Insulating materials are applied to the electrical machine in a variety of forms and materials.

Insulation types and variants in low-voltage machines < 1 kV



1. Wire insulation
 2. Groove base insulation
 3. Groove separation
 4. Cover slide
 5. Phase separation
 6. Cable insulation
 7. Bandage insulation
 8. Bandage insulation
 9. Cable insulation
 10. Wrap-around head bandage
 11. Bandage insulation
- Without number: Impregnation

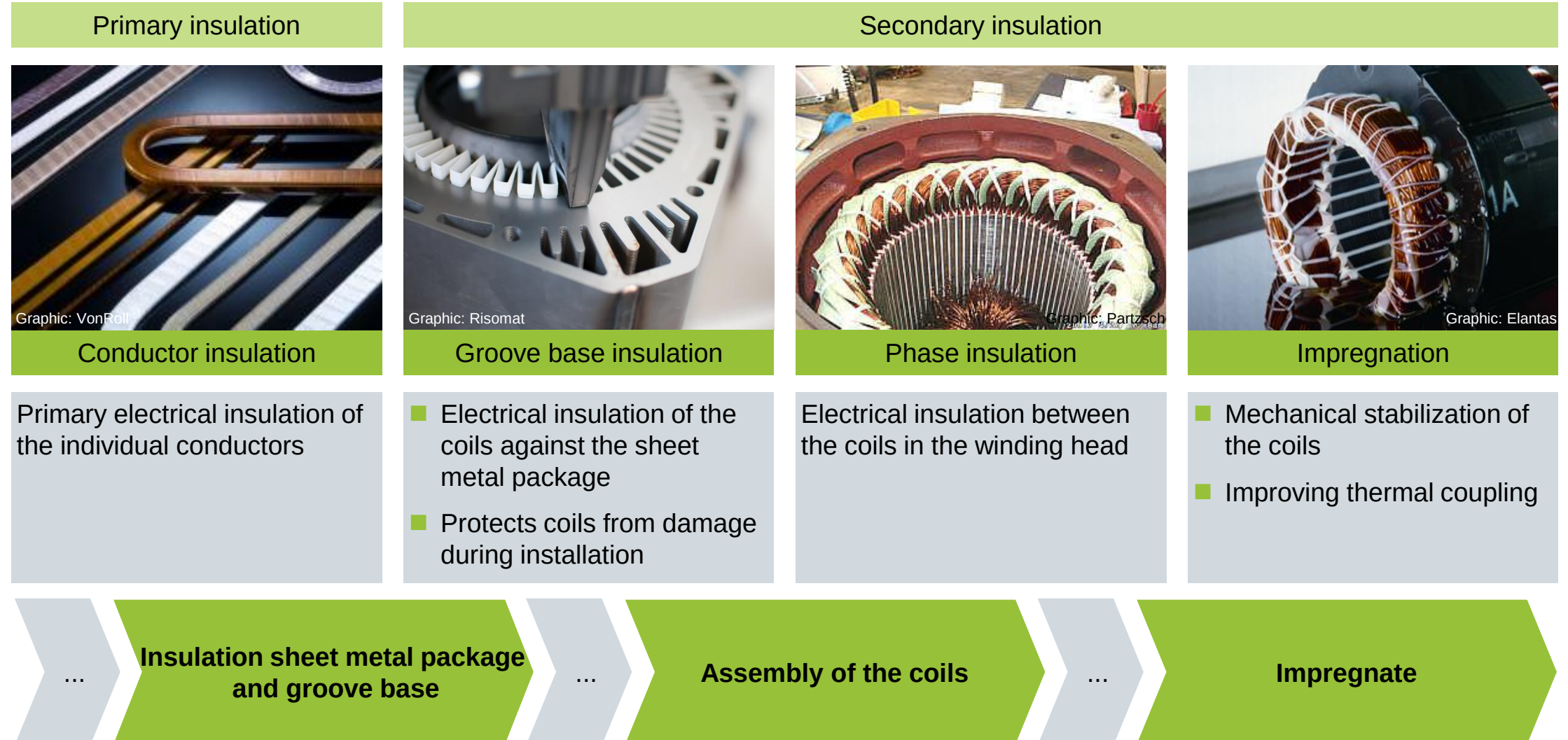
Primary insulation

Insulation layer that directly encloses the electrical conductor (wire enamel, sheet metal coating, bandages)

Secondary insulation

Additional insulation materials that are added during the assembly process (impregnating resin, insulation paper, groove separating slider)

The classic insulation system of a stator consists of several multifunctional levels.



The content of the lecture unit is intended to provide a basic overview of the relevant insulation steps involved in the manufacture of an electric motor.

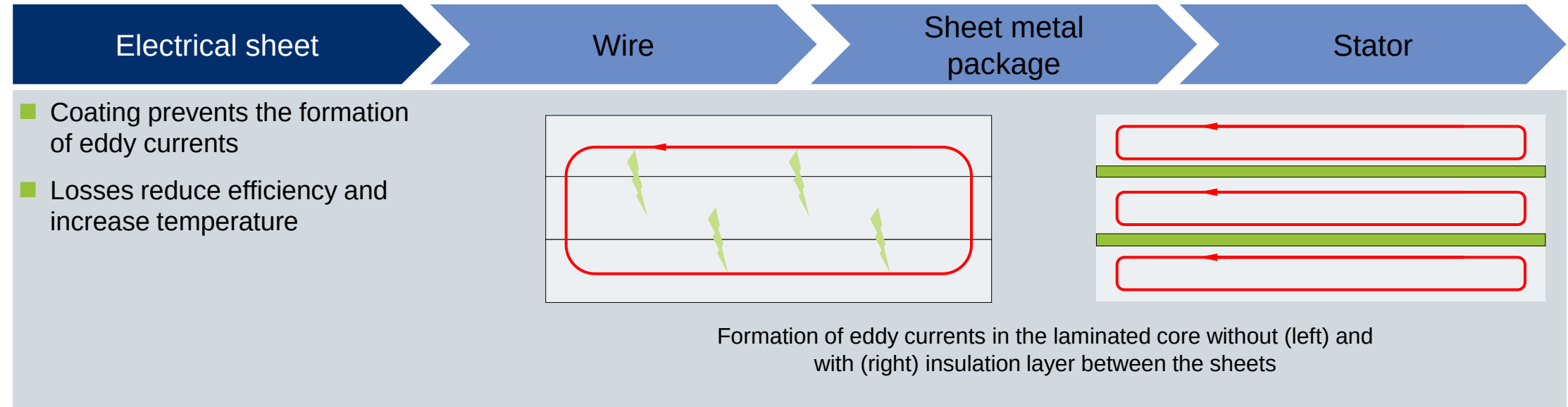
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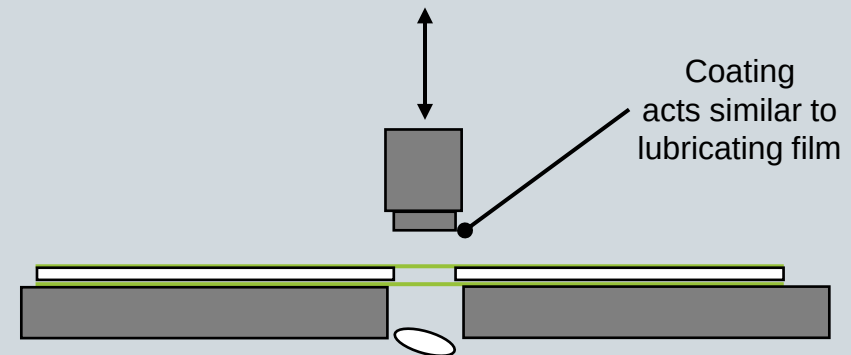
Graphic: azom.com

In addition to electrical insulation, the insulation of the sheet metal package also provides corrosion protection.

Slide from Lec. 2



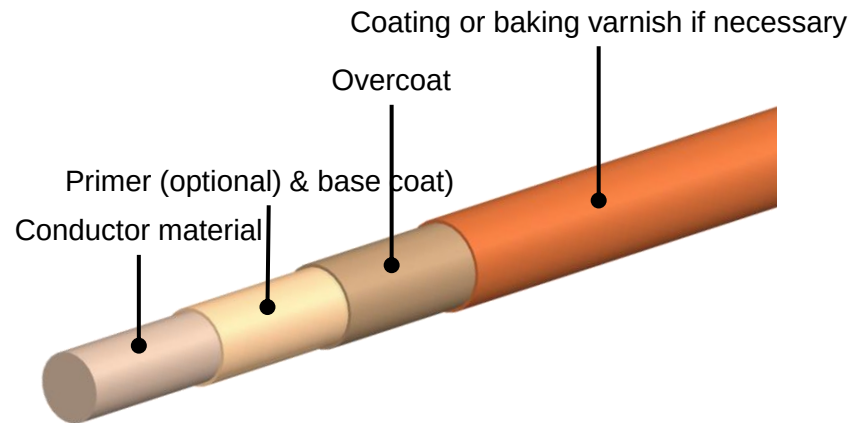
- Pre-applied insulating layer acts as a lubricating film during punching
- Improves cutting properties of otherwise brittle electrical sheet metal
- Increases tool life
- Protects against short circuits caused by burrs



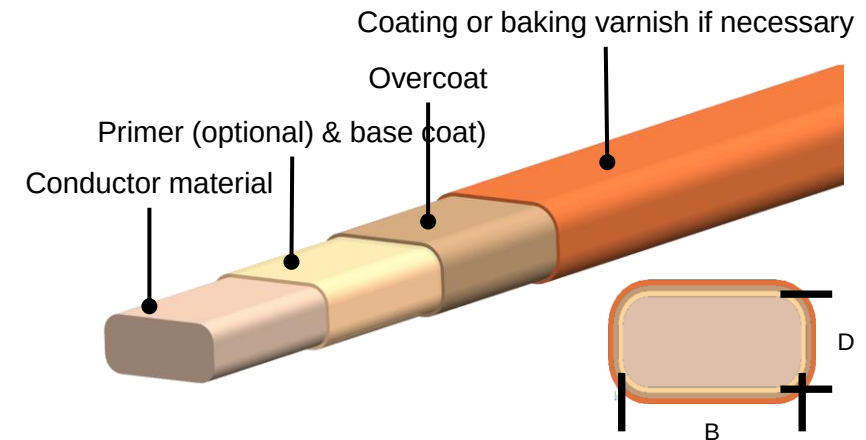
The structure of an enameled electrical conductor in winding technology differs from the material only in its geometry.

Slide from Lec. 4

Enameled insulated round conductor



Enameled insulated flat wire



- The carrier material can consist of different materials
- A base coat is applied over the carrier material to ensure defect-free application of the electrical insulation
- After the base coat, the electrical insulation layer (overcoat) is applied
 - Overcoat consists of synthetic resins and/or organic solvents
 - Curing takes place through a baking process
- The insulation is then supplemented with functional layers, such as an adhesive coating or a lubricating layer

The insulating varnish is applied in over to thirty layers depending on the desired thickness on the respective conductor.

Coating process

A distinction is made between two different procedures differentiated:

- Nozzle coating application
- Lacquer application with felt
 - 12 - 24 layers for thinner wires (0.012mm to 0.03mm)
 - 15 - 30 layers for medium wires (0.04mm to 0.15mm)
 - 12 - 24 layers for thicker wires (0.15mm to 5mm)

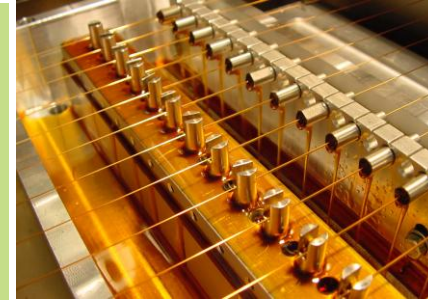


Structure of the coating system depending on the wire diameter:

- Vertical enameling system for larger wire diameters and profile wires
- Horizontal enameling system for thin wires

Classification of wire insulation according to DIN IEC 60317:

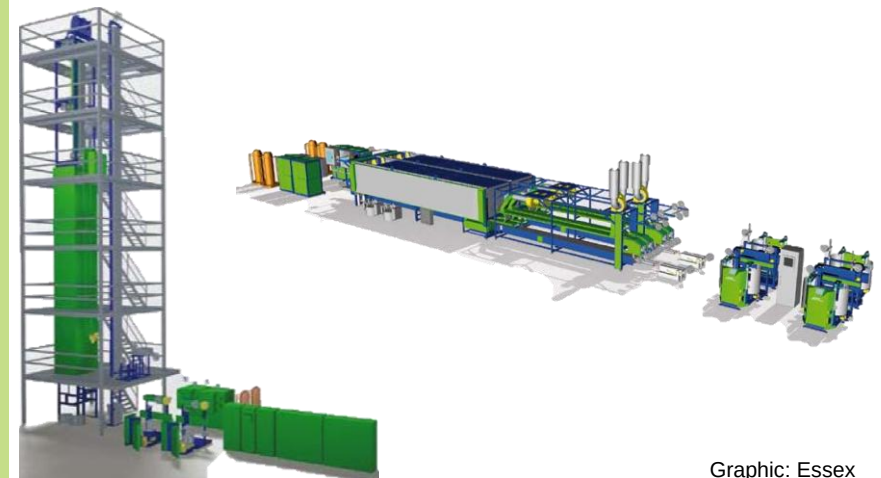
- Grade 1:
 - ~ 60 μm for flat conductors / round wires according to table
- Grade 2:
 - ~ 120 μm for flat conductors / round wires according to table
- Grade 3:
 - ~ 180 μm for flat conductors / round wires according to table



Graphic:
Aumann



Graphic:
Aumann



Graphic: Essex

As varnish insulation coatings are not suitable for every area of application, materials from the textile industry with various additives are also used.

Generally speaking, varnishes differ as follows:

- Solderable varnish, such as polyurethane
- Heat-resistant varnish, such as polyester
- Solvent-resistant varnish, s such as polyesteramidimide
- Baking varnishes, such as polyesteramide

Other insulation materials are

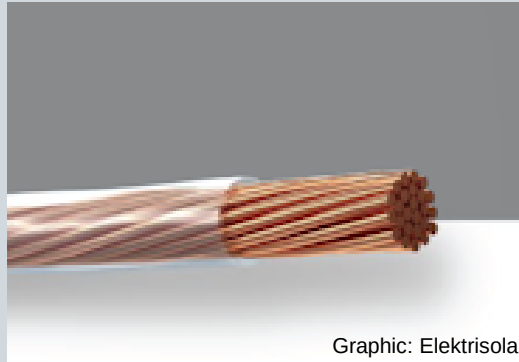
- Nylon
- Textile insulation for wires, such as E-glass yarn (alkali-free and impregnated with varnish)
- Tape and foil insulation, such as mica / polyester film or tape with epoxy resin

Material	Temperature index	Thermal shock	Softening temperature	Tin plating	Chemical resistance
Polyvinyl acetal	120°C	155°C	>215°C	No	Very good
Polyurethane	180°C	200°C	>250°C	Yes	Good
T. Poly. (imide) + polyamideimide	200°C	220°C	>340°C	No	Good
Polyamideimide	220°C	240°C	>400°C	No	Very good
Polyimide	240°C	260°C	>500°C	No	Very good

Source: Essex

In addition to enamel-insulated conductors, there are a variety of other primary insulations with different structures.

Extruded



- Seamless wrapping of the conductor material by means of plastic extrusion
- Excellent chemical and mechanical resistance
- Typical materials: polyester, polyamides, polyether ether ketones (PEEK)

Bandaged



- Single or multi-layer taping with variable degrees of overlap
- High dielectric strengths and temperature classes possible
- Typical materials: polyimide, polyester

Braided



- Multi-layer coating of the conductor material with yarn
- Good form stability, flexibility and impregnability for stranded conductors
- Typical materials: polyester, nylon, natural silk

Anodized



- Passivation of the surface by energizing the conductor material in an acid bath
- Mainly used for aluminum materials (less frequently Cu)
- Use with stranded conductors in combination with additional wrapping

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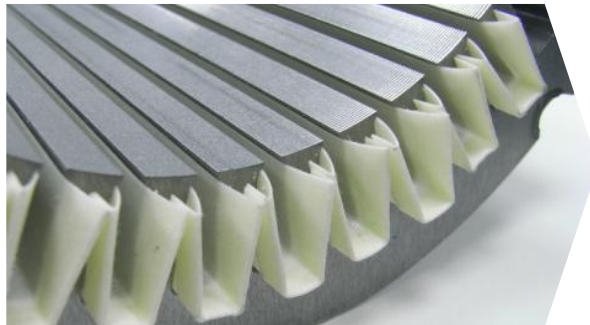


Graphic: azom.com

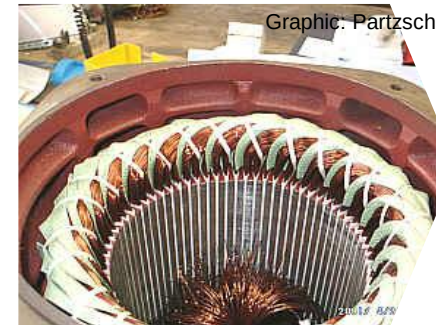
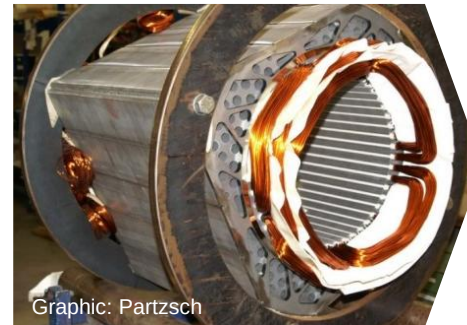
The insulation is applied to the motor at various stages of the production process.

Insulation-relevant individual steps (using the example of the process chain of a wound stator / rotor)

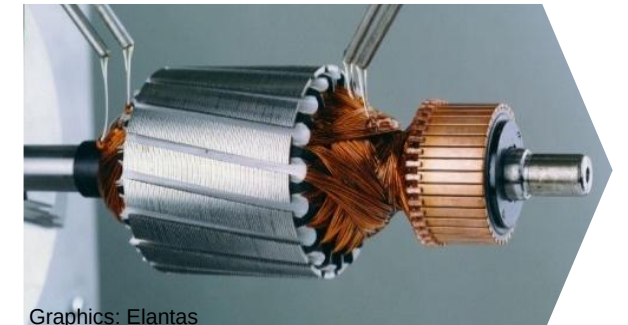
- Pulling paper insulation into groove base /
- Application of powder coating



- Enamel insulation is already part of the delivered wire
- In the winding process, further insulation is applied for phase separation



Impregnation of the entire installed winding by dipping, trickling or pouring



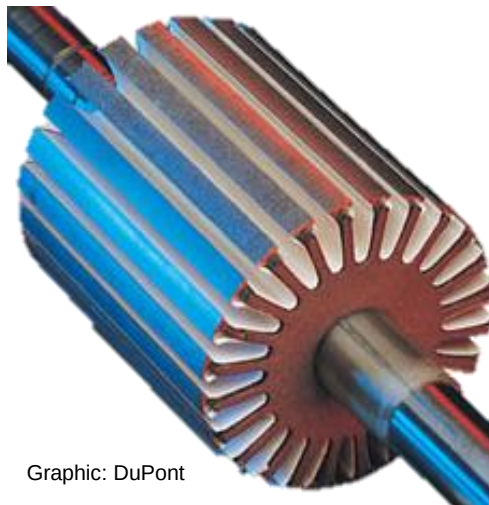
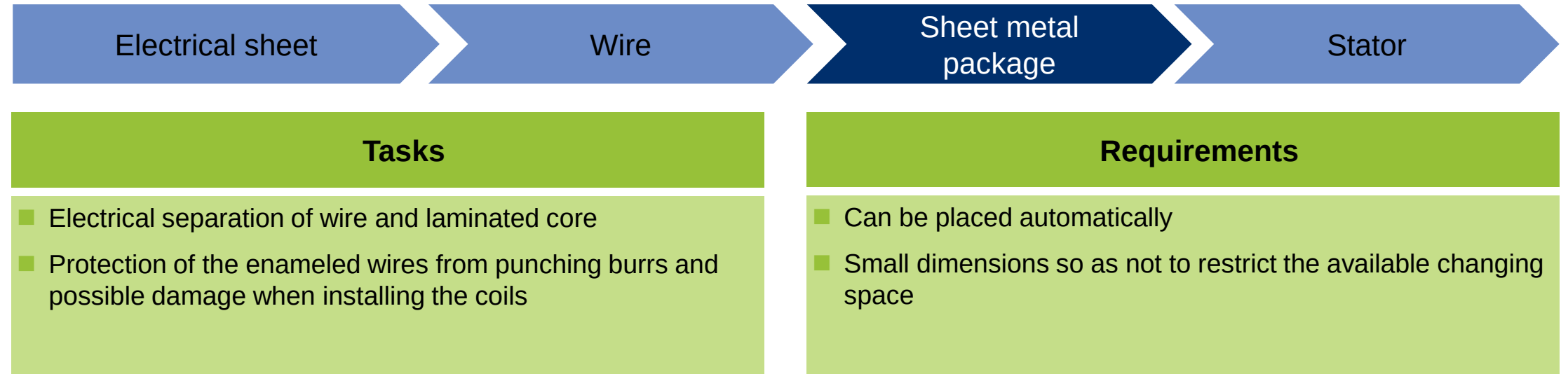
...
**Insulation sheet
metal package and
groove base**

...
**Mounting / Insulating
single coils**

...
**Phase insulation
winding head**

...
Impregnate

The slot base insulation in the form of slot liners, powder coating or plastic parts separates wire and sheet metal from each other.



Procedures

Slot liner

Powder coating

Plastic films

Plastic molded parts



The automated insertion of slot liners is the most common method of slot base insulation for industrial machines.

Electrical sheet

Wire

Sheet metal
package

Stator



Graphic: DuPont

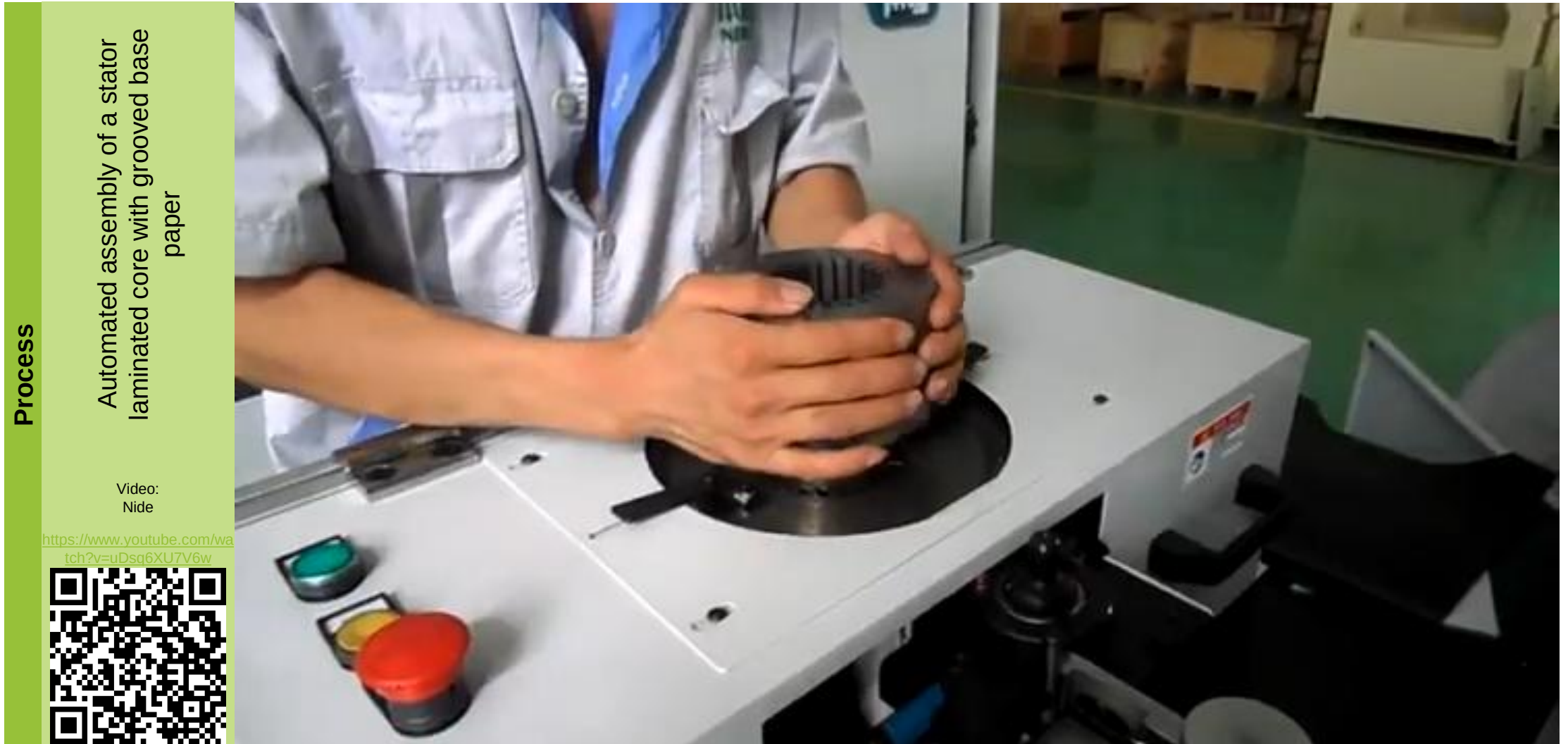
- Insulation paper is unwound directly from the roll as a semi-finished product and cut to length in the machine
- Cut-to-length pieces are automatically folded and inserted
- Rebate on both sides prevents slipping out of the stator

Materials

- Mica paper
- Aramid paper (Nomex)
- Plastic films

Area of application: Can be used in principle for all motor sizes, most frequently in industrial motors from 150 mm active length, as this is where it is easiest to use.

The automated insertion of slot liners is the most common method of slot base insulation for industrial machines.



The powder coating process is currently mainly used for flyer-wound small drives.

Electrical sheet

Wire

Sheet metal
package

Stator



Graphic: 3M

- Powder insulation can replace final impregnation for low-voltage machines (~ 48V)
- Functional surfaces are usually reworked or protected during powder application
- Application under rotation ensures uniform layer thickness
- Powders are cured in a downstream oven process

Materials:

Mostly powdered epoxies, or epoxy-polyester mixtures

Area of application: Mainly smaller, wound machines (windshield wiper motor,...) which can be heated up quickly (Curing process at 160-180°C usual)

The powder coating process is currently mainly used for flyer-wound small drives.

Process

Powder coating of rotor laminations



Compared to conventional inserts, powder coating can improve the windability of rotors and stators.

Specific properties of powder-coated, wound motor components

Advantages

- Uniform, dielectric strength insulation layer can be achieved
- Coating independent of surface geometry
- Simultaneous corrosion protection and insulation
- Bending radii of the wires are increased
- More compact motor, smaller winding head

Disadvantages

- Industrially highly automated only for smaller, wound motors (mostly DC brushless)
- Functional surfaces must be cleaned
- Lower cycle time compared to slot liners

Specific properties of Slot liner / paper insulation

Advantages

- Cost-effective and easy to automate
- High temperature resistance $> 200^{\circ}\text{C}$
- Good chemical resistance to aggressive media
- Insensitive to moisture
- Very fast process

Disadvantages

- Sharp bending radii for wrapped wires
- Suboptimal for needle winding processes
- Enlargement of the winding head

Additional injection-molded parts facilitate stator winding with distributed and concentrated winding.

Injection-molded end caps

- Insulation of the windings from the package at the ends using so-called insulating stars
- In combination with flexible groove lining, this is referred to as hybrid insulation



Graphics: PVS

Inner stator insulation using molded parts

- End insulating parts with axially projecting guide hooks, whose radial supports dock onto the end of the package
- Stator / rotor does not require slot base insulation made of paper



Complete overmolding

- Complete overmolding with or injection of thermoplastic
- Injection mold and pneumatically actuated cores ensure constant inner / outer diameter
- Use with complete stators and individual coils



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Graphic: azom.com

Final impregnation is a complex and critical step in the production of electrical machines.



Impregnation in the context of electrical engineering:

Impregnation describes the final complete impregnation of a wound stator or rotor laminated core with an additional insulating layer consisting of a liquid insulating resin (usually thermoset), which then hardens thermally.

Why?

- Impregnating resins can fill any damage caused during the winding or drawing-in process of the windings and restore the insulation effect
- The extremely strong thermosetting resins give the winding head mechanical stability after curing
- The insulation layer also has other positive properties for operation of the machine

Which materials?

- Thermosetting plastics, i.e. plastics that do not melt again, which harden into molding compounds through a chemical reaction
- Mainly in the production of electric machines:
 - Epoxy resins
 - Polyester resins / polyesterimides
 - Polyurethane resins
 - Silicone resins

In addition to electrical insulation, the final resin insulation performs other functions within the electrical machine.

Priority for the configuration of the insulation

Electrical insulation

Electrical separation of:

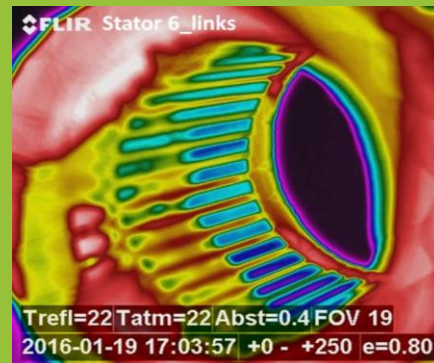
- Individual sheets
- Wires in relation to each other
- sheet stack and wire winding as a whole



Graphic: Elantas

Thermal coupling

- Thermal connection between wire, laminated core and housing
- Improved heat dissipation from the winding



Increase in strength

- Mechanical stabilization of the winding head with distributed windings
- Fixing the winding within the grooves



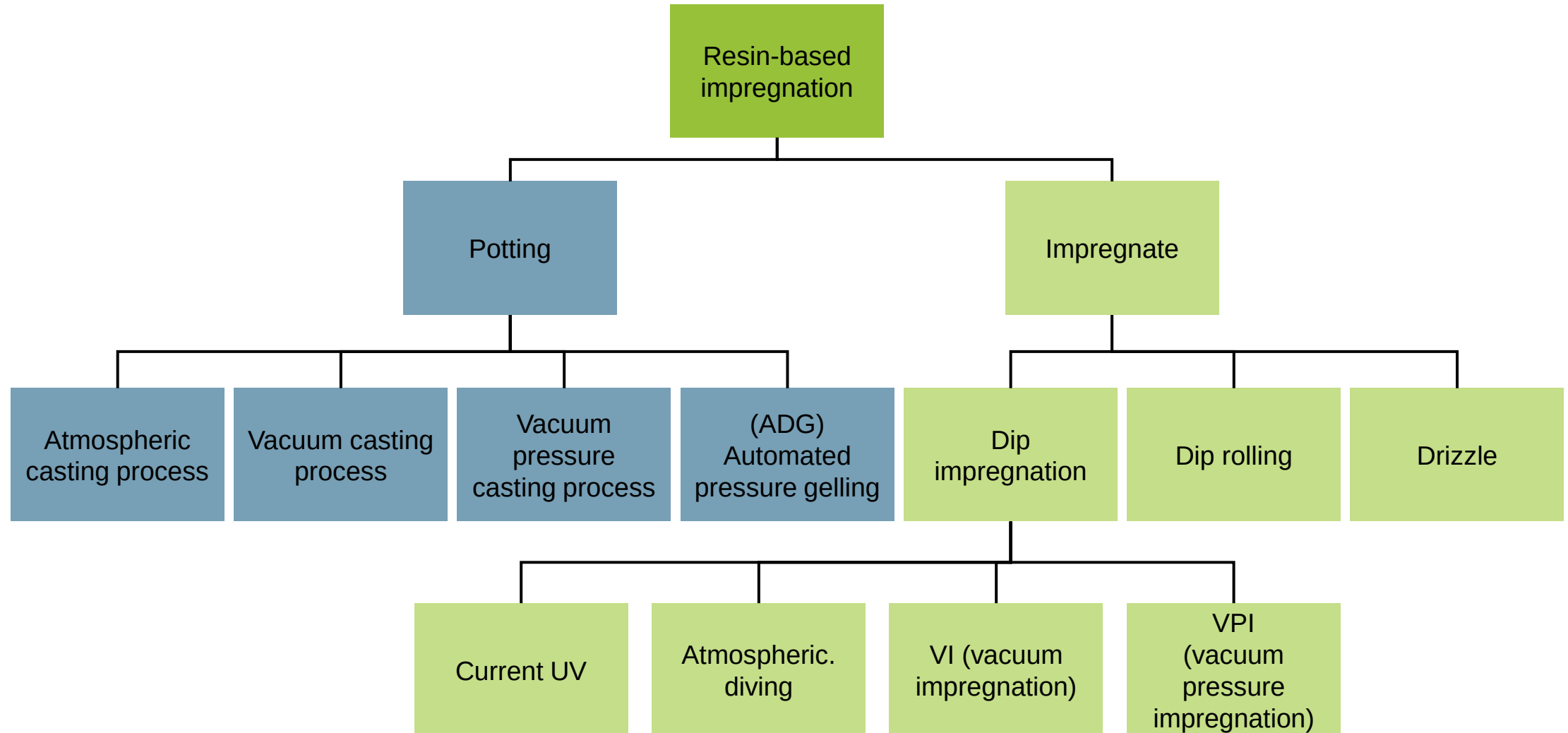
Encapsulation

Separation of sensitive components such as wire and magnets from aggressive media such as oil, water or other operating fluids



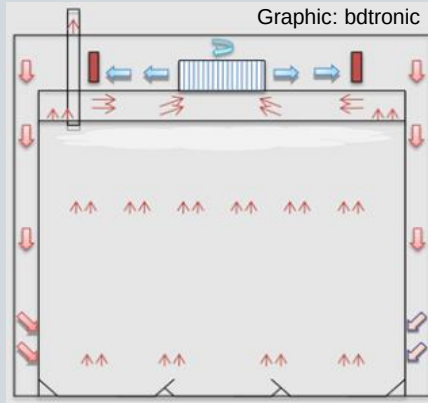
Graphic: Elantas

The application methods for impregnating resins are many and varied, depending on the type of machine and application.



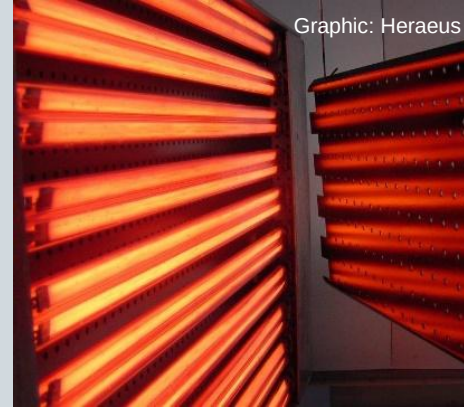
The different curing processes have an impact on the costs and economy of the production process.

Convective



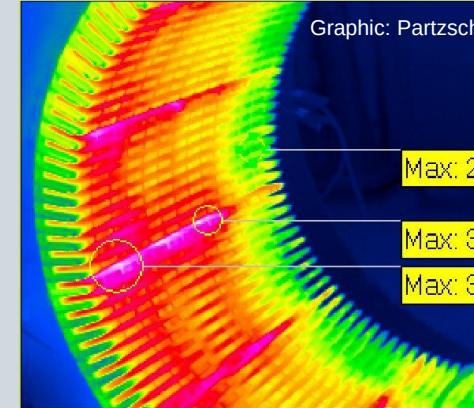
Convective transfer of energy with air as the transfer medium

Infrared



Parts are exposed to near infrared light, energy transfer through radiant heat

Joule effect



Energizing the windings with direct or alternating current causes resistance heating

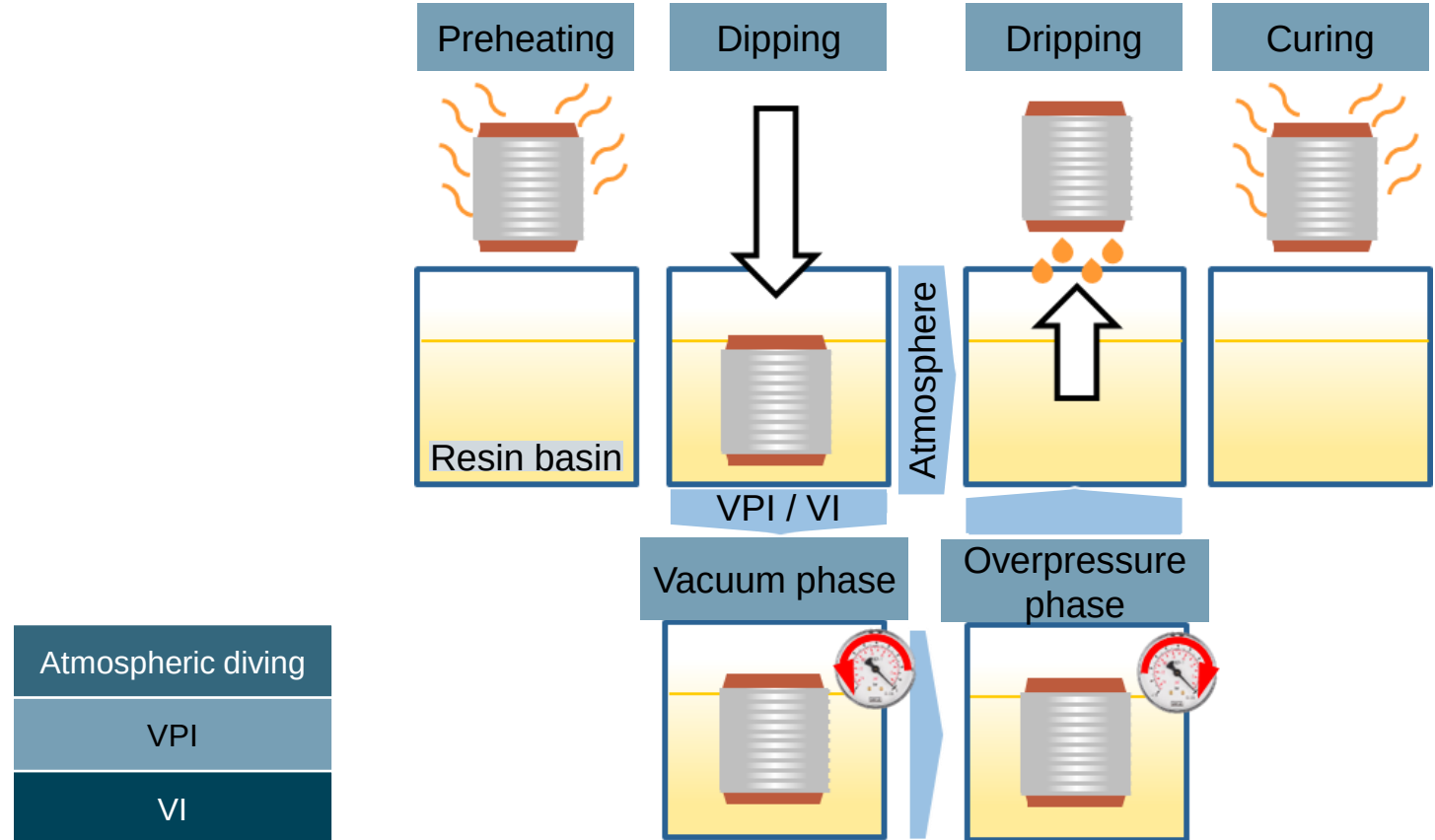
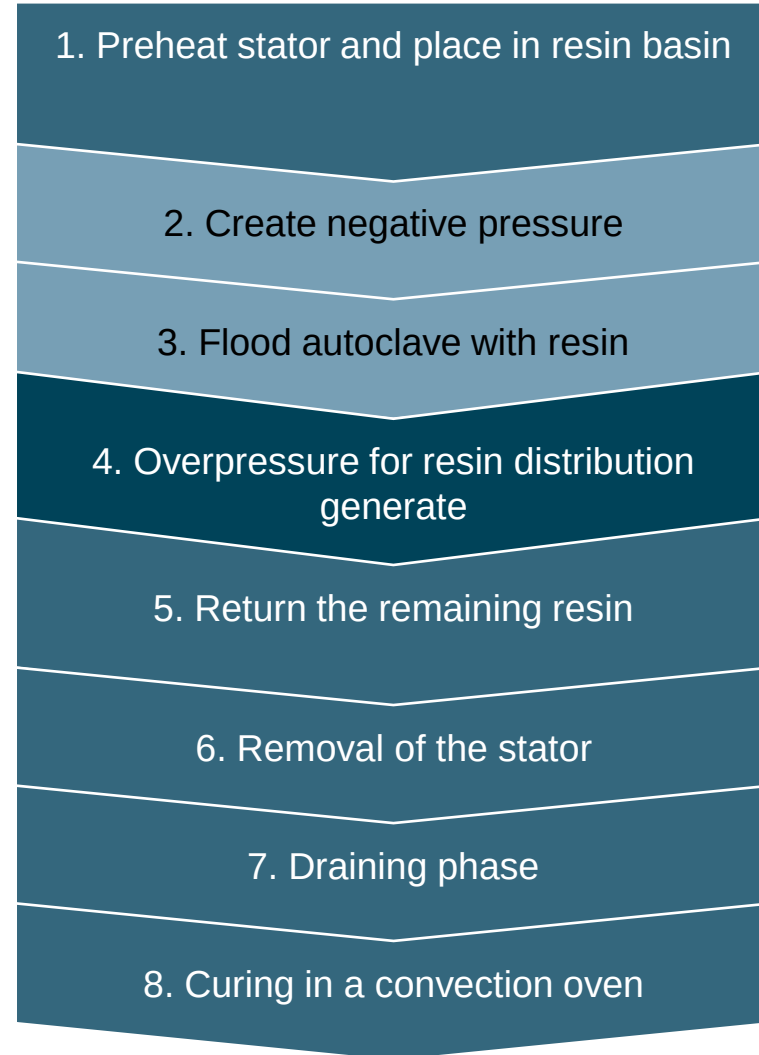
Inductive



Inductive heating by generating eddy currents in the component

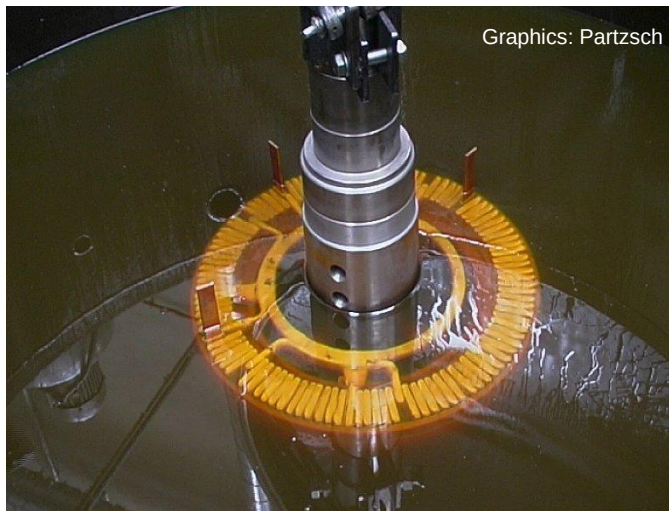
Advantages	- Uniform heating - Hermetically sealed	Direct heating of the components, surroundings remain cold	Lower power consumption (70% compared to oven)	- Very fast heating - Wires are not stressed
Disadvantages	- Inefficient ($\eta=40\%$) - Significant energy consumption	- Low penetration depth - Inefficient for complete heating	- Load on the windings - thermal ageing of the dielectric	Little experience with use in the production of electric machines

Atmospheric / pressurized dip impregnation is the most cost-effective and least complex impregnation method.



- Atmospheric dipping for machines with low loads (< 400V)
- Optional preheating in the oven increases the resin filling level
- VI / VPI common for high-voltage machines (from 1kV) or generators

Atmospheric / pressurized dip impregnation is the most cost-effective and least complex impregnation method.



Advantages of dip impregnation

- Great flexibility with regard to the stators to be impregnated
- Maximum size limited only by tank
- Comparatively low equipment costs

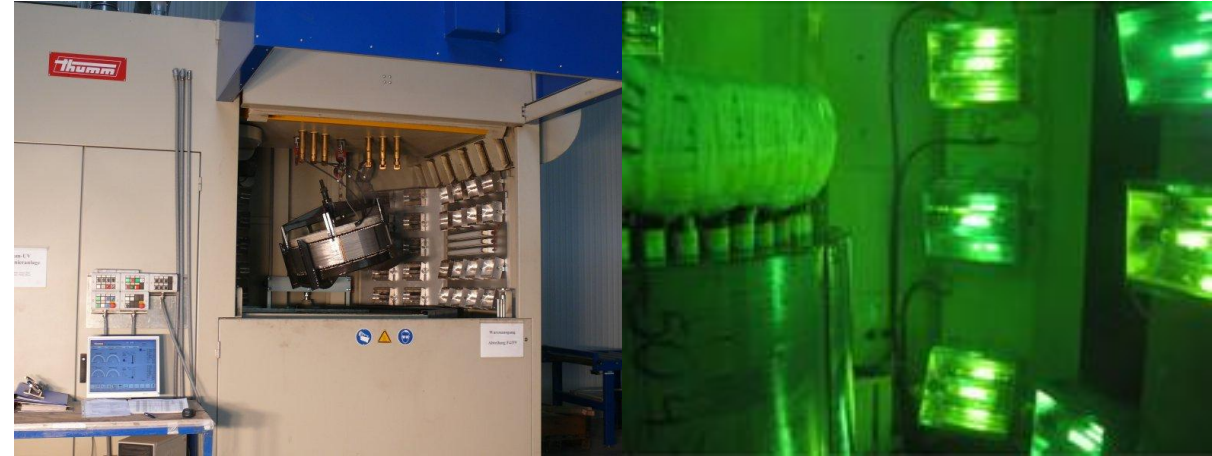
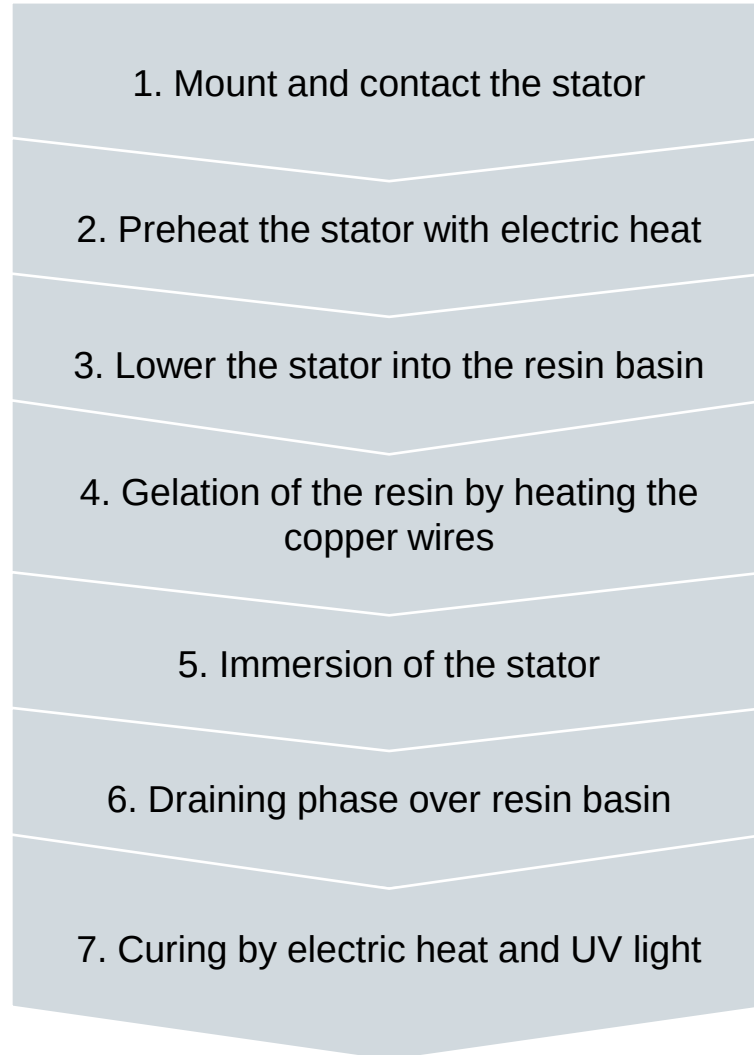
The following also applies to vacuum pressure impregnation:

- Optimum material feed into the winding material and high filling levels possible
- Processing of materials with different viscosities

Disadvantages

- High material losses due to dripping
- Problematic with non-solvent-free materials
- Rising system costs due to vacuum pumps and overpressurization
- Relatively long process times per component (even higher with VPI)

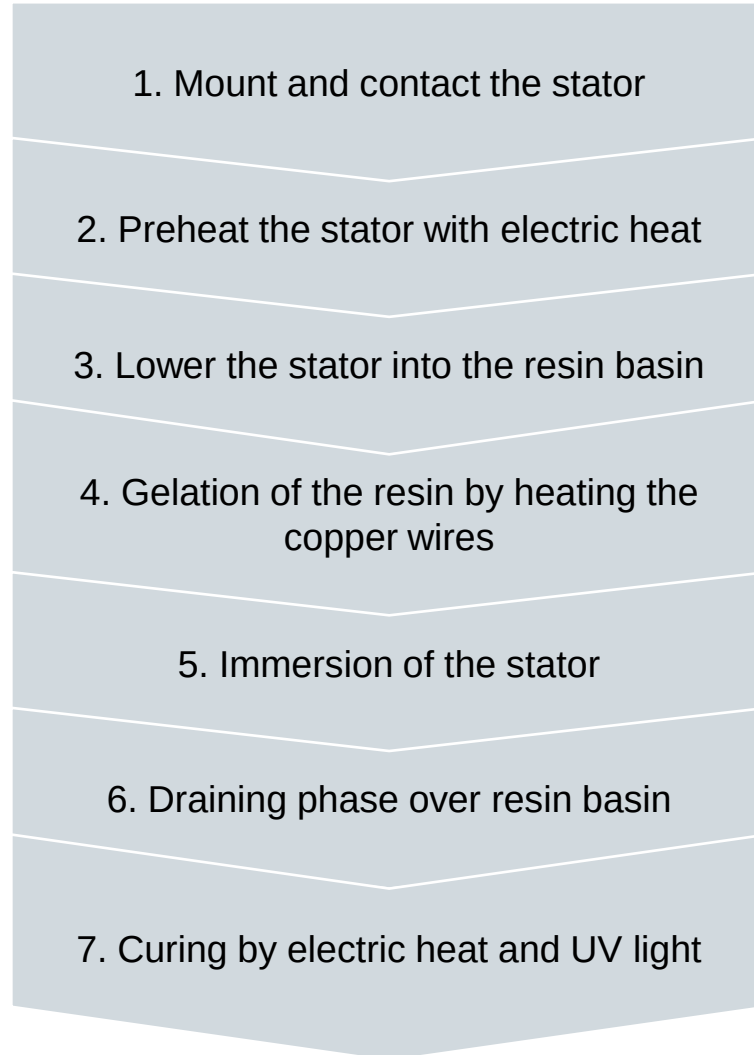
The current UV process provides fast and high quality impregnation of machines.



Graphics: Partzsch / Elin EBG

- Use of special UV impregnating resins that can be cured using UV light
- Heating takes place via the contacted windings
- The stator is immersed in hot water. After the holding phase, the resin on the surface is cured immediately by large area UV irradiation.
- Residual resin is cured by further heating
- Common for low-voltage and high-voltage machines up to medium size

The current UV process provides fast and high quality impregnation of machines.



Advantages

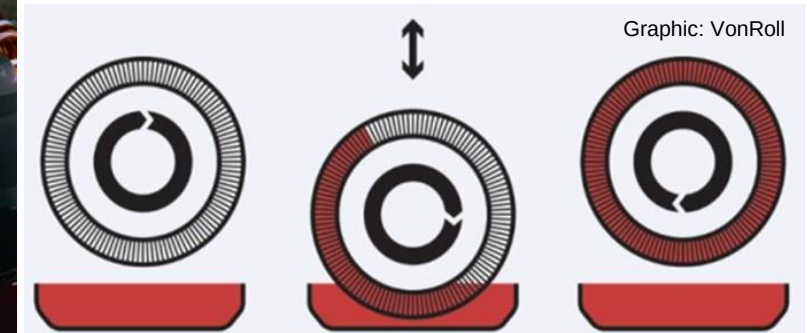
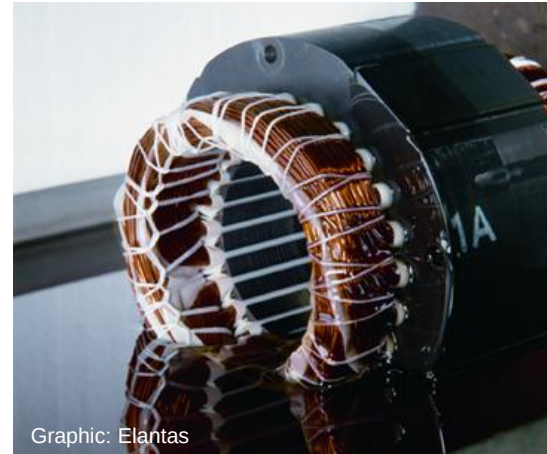
- Very good impregnation results with a high filling level
- Low drip losses
- Short processing times
- Reduced post-processing (drip-off noses)
- Lower energy consumption by eliminating oven curing
- Low emissions
- Suitable for automated production process

Disadvantages

- High equipment costs, complex system technology
- High-energy UV light requires encapsulation to protect employees
- High thermal load on the windings can cause the dielectric to age prematurely

Dip rolling offers a low-rework alternative to dip impregnation.

1. Clamp the stator horizontally in the device
2. Preheating under roller burnishing
3. Lifting of the resin basin
4. Impregnation under electric heat and roller burnishing
5. Lowering of the resin basin
6. Draining phase
7. Gelling phase
8. Curing in the oven

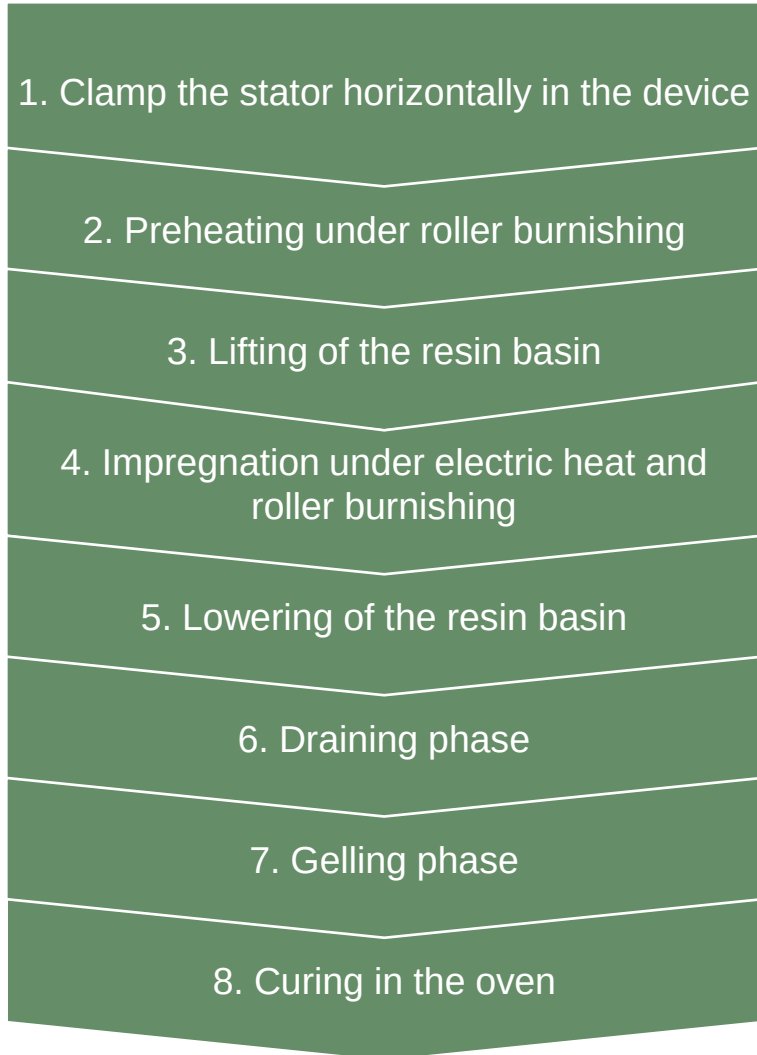


- The rotor to be impregnated is partially dipped in the resin and rotated
- Only the windings and slot base are in the resin bath
- (Pre-) heating by oven or electrical heat possible
- Curing of the resin also takes place under constant rotation.
- Common for low and high voltage machines, including larger diameters

Dip rolling offers a low-rework alternative to dip impregnation.



Dip rolling offers a low-rework alternative to dip impregnation.



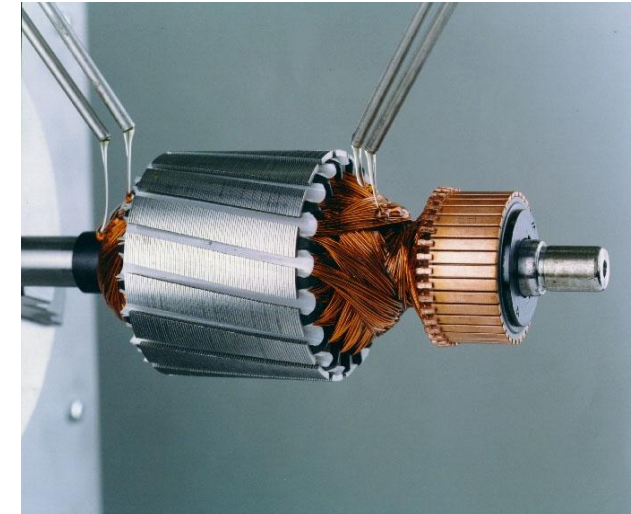
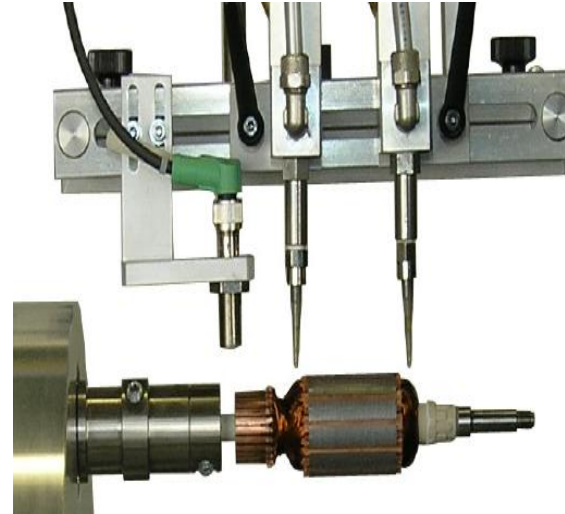
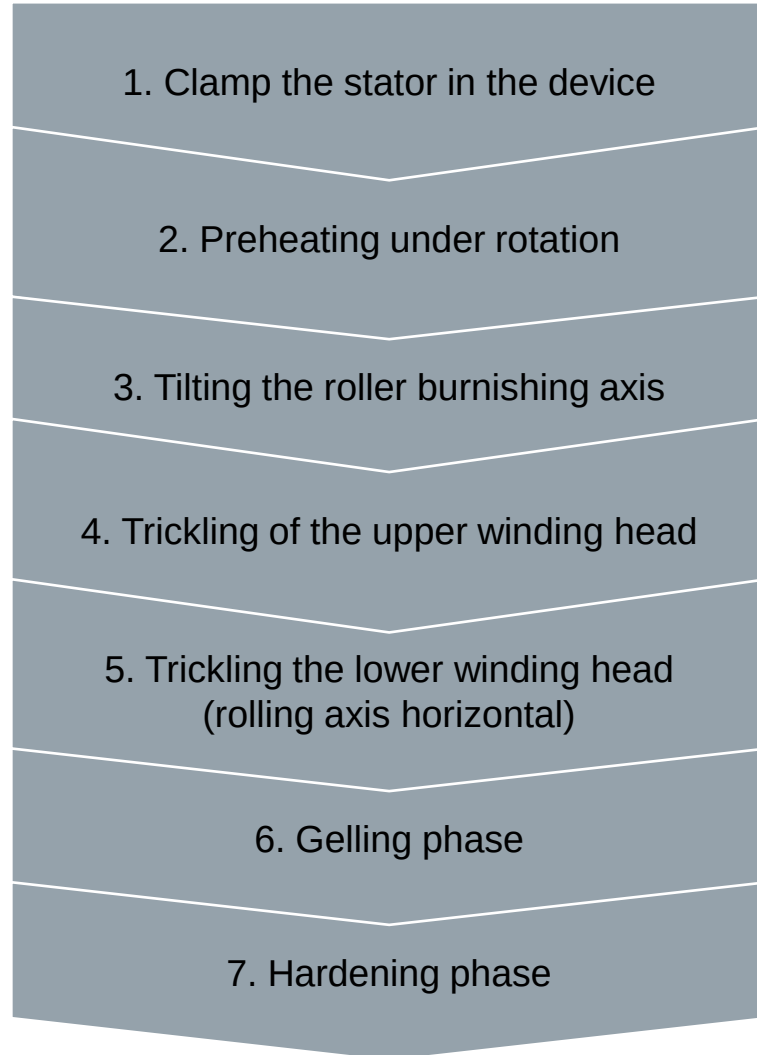
Advantages

- Rotation ensures even resin distribution in the stator / rotor
- Less material loss (dripping) than with dip impregnation
- Suitable for the automated production process
- High material input into the winding material
- Inexpensive and simple procedure

Disadvantages

- Resin filling level in the groove base may be insufficient
- Open storage in the basin promotes ageing / contamination
- External surfaces must still be reworked
- Limited flexibility with regard to winding material sizes

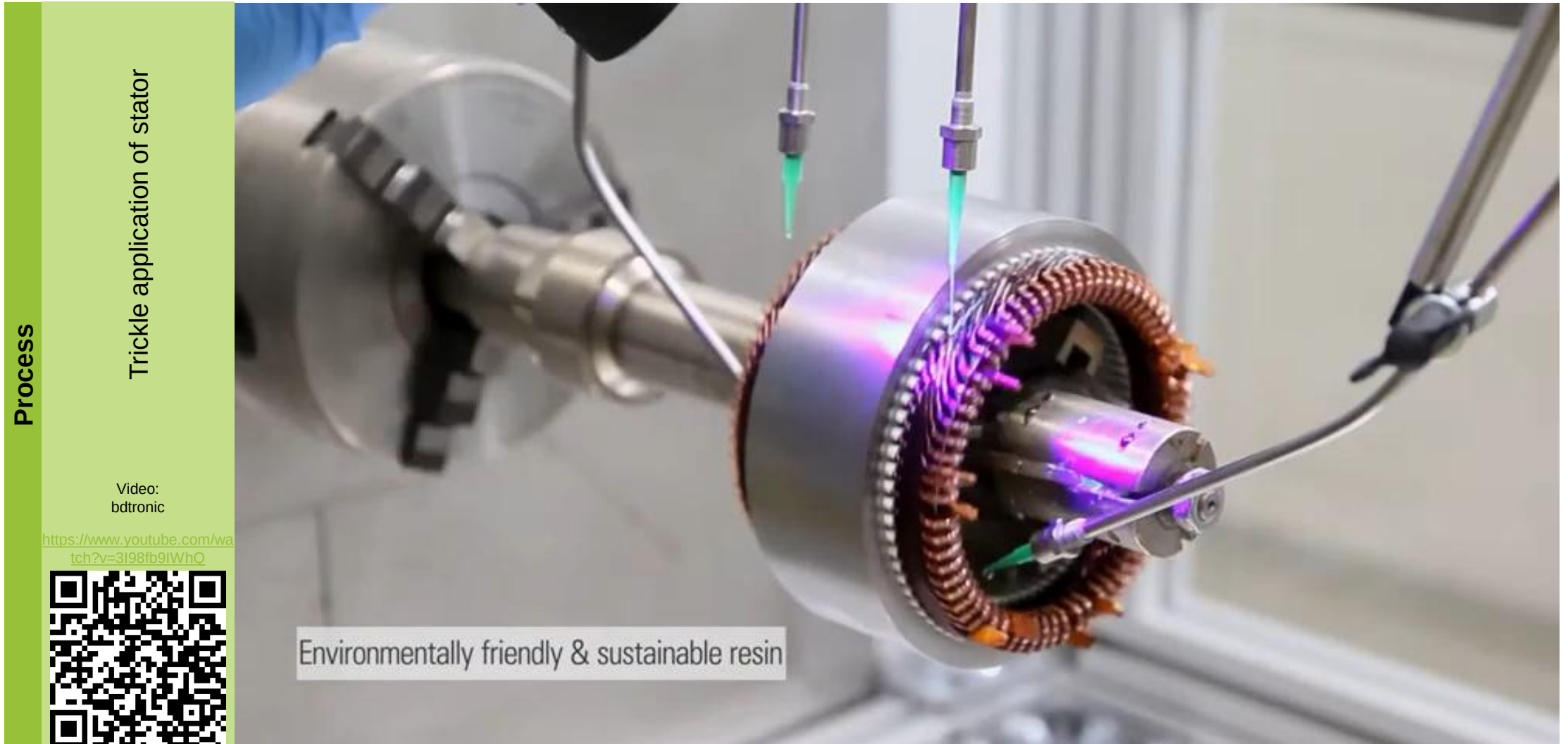
The trickle impregnation achieves almost 100% material utilization for smaller motors.



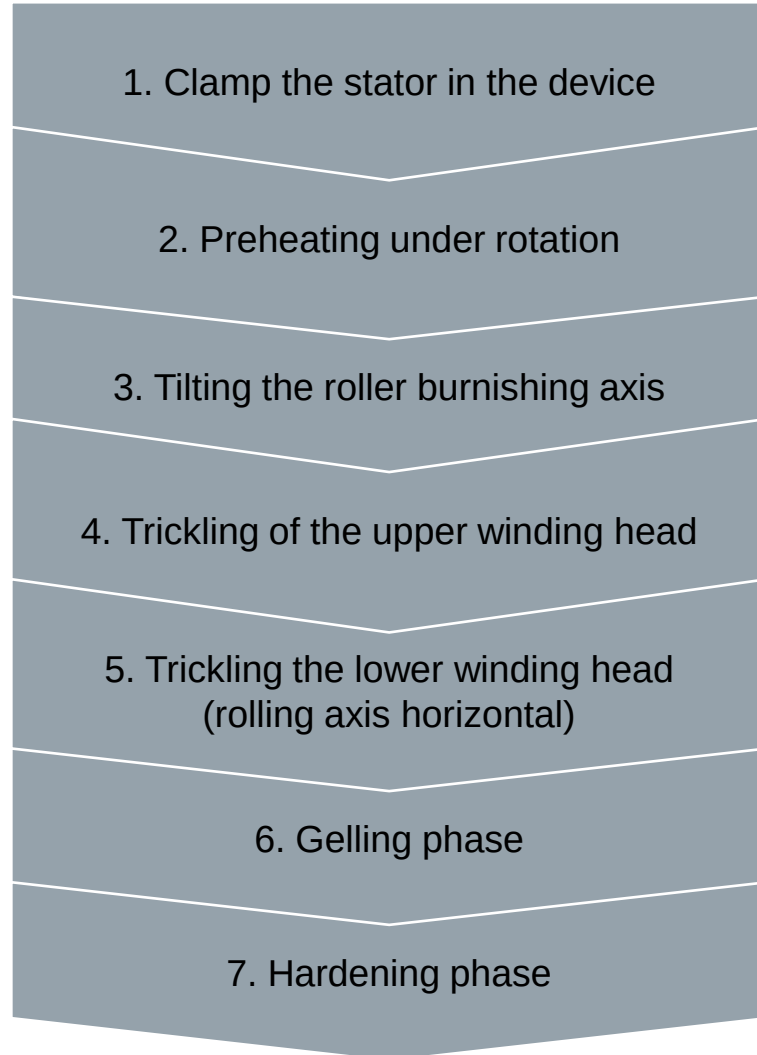
Graphics: bdtronic, Elantas

- Rotor to be impregnated is set in rotation and resin is fed directly onto the windings from nozzles
- Capillary action causes soaking of the windings
- Curing of the resin also takes place under constant rotation
- Common with compact rotors or internal stators (can also be easily automated in this cases)

The trickle impregnation achieves almost 100% material utilization for smaller motors.



The trickle impregnation achieves almost 100% material utilization for smaller motors.



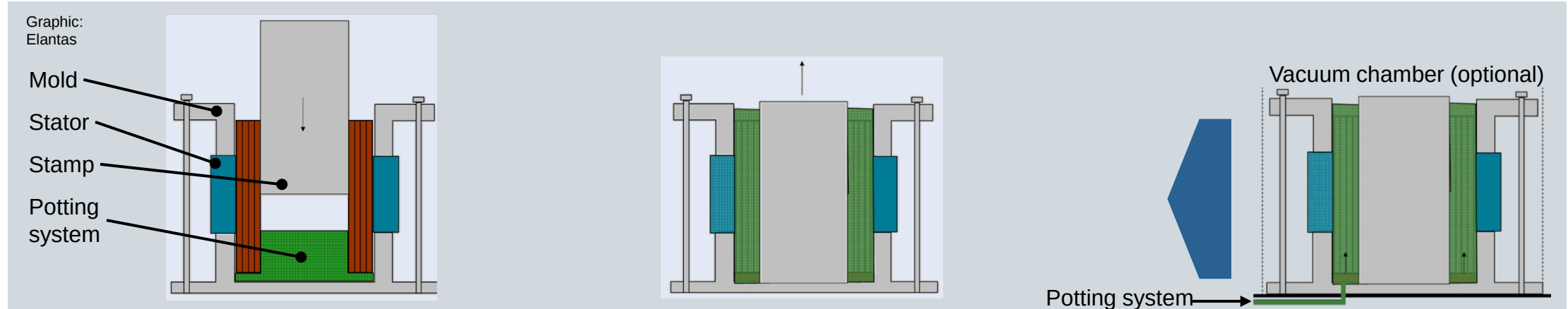
Advantages

- Excellent material utilization as there is practically no drip loss and no need to store material in basins
- Suitable for automated production
- Short cycle times and high volumes possible
- Can be combined with electrical heating or oven curing

Disadvantages

- Optimum resin application only with tilting of the winding material
- Low flexibility of the system technology, for large-scale production of identical parts
- Complex system technology
- Less suitable for formed coils due to lack of capillary effect

Atmospheric potting makes it possible to significantly reduce the complexity of the system, but requires movable molded parts.



- Stator is integrated into sealed mold prior to start
- A predetermined volume of resin is poured into the center of the mold.
- Punch is lowered by machine (fixed feed rate)
- Isostatic pressure forces resin into windings against punch feed
- When using a 2-component resin, the core is removed from the mold after the cure time.
- Alternatively, encapsulation can be performed under vacuum

Advantages

- High filler content up to 70% processable
- Significant improvement in electrical insulation and heat conduction
- Variants for potting under vacuum
- Constant resin feed (stamp)

Disadvantages

- Bubble-free potting depends on correct stamp feed
- Demolding the stamp
- No widespread industrial application yet; More weight

Each of these methods offers different advantages and disadvantages depending on the task at hand.

	Dipping	Current UV	VPI	Dip-rolling	Trickling	Potting
Filling level	< 50%	> 90%	up to 90%	50-80%	>90%	100%
Automation capability	0	0	-	++	++	0
Process time	--	+	--	+	+	+
Minimal post-processing	--	0	--	-	++	++
Costs	low	high	medium	low	low	high

The content of the lecture unit is intended to provide a basic overview of the relevant insulation steps involved in the manufacture of an electric motor.

- Basics of electrical insulation systems
- Materials and types of insulation systems
- Process for applying the primary insulation
- Method for applying the secondary insulation
- Resin-based impregnation and potting processes
- **Failure modes and testing technology**



Graphic: azom.com

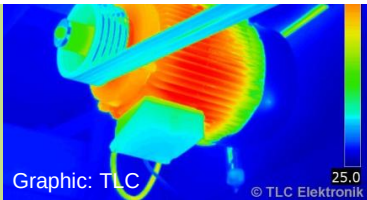
The causes of insulation damage are varied and can also overlap.



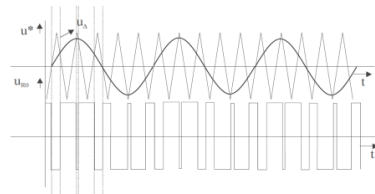
Overvoltage

Electrical breakdown when the limiting field strength of the dielectric is exceeded

Thermal Load



Thermal runaway due to insufficient heat dissipation



PWM operation

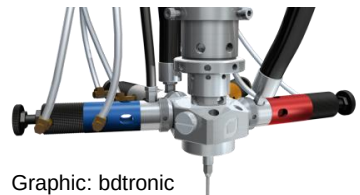
High pulse frequency: Voltage gradients cause polarization losses and large potential differences

Failure mechanisms of a solid dielectric depending on exposure time and field strength

Environment variables



Chemicals, dirt, abrasion, moisture



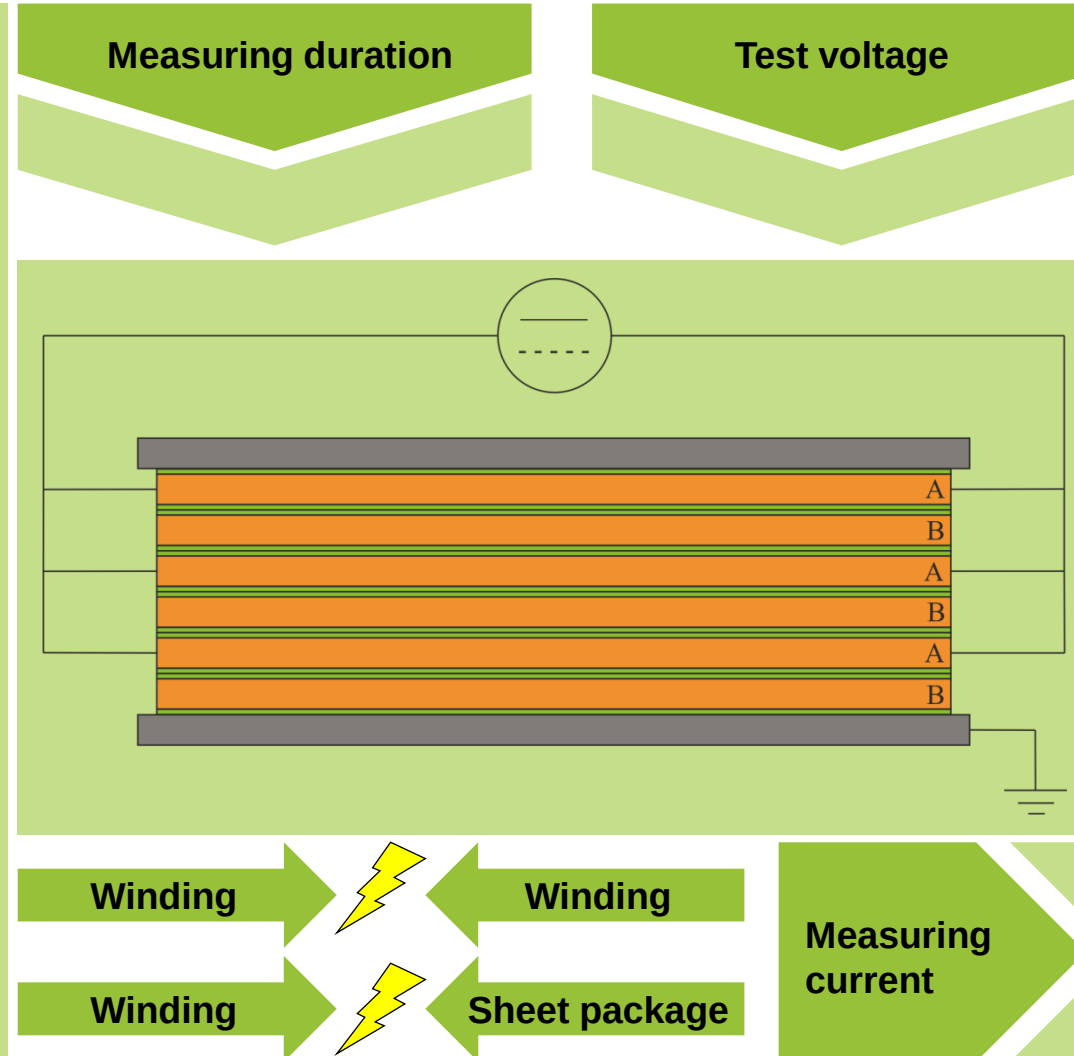
Processing

Mechanical stress, e.g. from winding process, cracks/defects:
Ignition source for partial discharge erosion resulting in failure



Various measurements are used to check the insulation properties of wound stators.

- Insulation resistance measurement
- High voltage test
- Protective conductor resistance measurement
- Resistance measurement
- Surge voltage test
- Partial discharge measurement
- Polarization index measurement
- Loss angle measurement
- Temperature measurement/ thermography
- Performance test with built-in rotor



- e.g. 500 V DC direct voltage according to DIN VDE 0100-600
- No influence with DC voltage due to small capacitances of the insulation material
- For large machines (large capacities), longer measurement is still necessary to ensure that the high voltage is present
- High insulation resistance is an indication of good insulating capacity

The contents of the lecture unit can be further deepened with the following specialized literature:

Reference books

- Küchler, A. (2017): High voltage technology. Basics - Technology - Applications. 4th edition. Berlin, Heidelberg: Springer Berlin Heidelberg (VDI book).
- Hagedorn, J.; Sell-Le Blanc, F.; Fleischer, J. (2016): Handbook of winding technology for high-efficiency coils and motors. Berlin, Heidelberg: Springer Berlin Heidelberg.
- Gherardi, P.; Sulzbach, H. (2008): Polymers for electrical insulation. Coatings and casting materials for the electrical industry. Landsberg, Lech: Publ. Moderne Industrie (Die Bibliothek der Technik, 314).
- Wei Tong (2014): Mechanical Design of Electric Motors. Boca Raton, London, New York. CRC Press





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THANK YOU